A Survey of Personalization: From RAG to Agent

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Personalization has become an essential capability in modern AI systems, enabling customized interactions that align with individual user preferences, contexts, and goals. Recent research has increasingly concentrated on Retrieval-Augmented Generation (RAG) frameworks and their evolution into more advanced agent-based architectures within personalized settings to enhance user satisfaction. Building on this foundation, this survey systematically examines personalization across the three core stages of RAG: pre-retrieval, retrieval, and generation. Beyond RAG, we further extend its capabilities into the realm of Personalized LLM-based Agents, which enhance traditional RAG systems with agentic functionalities, including user understanding,

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personalized planning and execution, and dynamic generation. For both personalization in RAG and agent-based personalization, we provide formal definitions, conduct a comprehensive review of recent literature, and summarize key datasets and evaluation metrics. Additionally, we discuss fundamental challenges, limitations, and promising research directions in this evolving field. Relevant papers and resources are continuously updated at the Github Repo (https://github.com/Applied-Machine-Learning-Lab/Awesome-Personalized-RAG-Agent).

CCS Concepts: • **Information systems** → **Personalization**;

Additional Key Words and Phrases: Agent, Large Language Model, Personalization, Retrieval-Augmented Generation

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1 Introduction

Large Language Models (LLMs) have revolutionized AI-driven applications by enabling natural language understanding and generation at an unprecedented scale. However, these models often suffer from issues such as outdated responses and hallucinations, which severely hinder the accuracy of generated information. **Retrieval-Augmented Generation (RAG)** has emerged as a promising framework that integrates retrieved information from external corpora, such as external APIs [16, 47], scientific repositories [119, 161], or domain-specific databases [7, 40], ensuring outputs that are more grounded in knowledge and up-to-date.

The versatility of this framework has led to significant applications across various domains, including question answering [151], enterprise search [19], and healthcare [183]. Among these applications, one particularly notable area is the workflow of agents, where RAG enhances autonomous systems by providing context-aware, dynamically retrieved, and reliable knowledge. This enhancement occurs because each stage of the RAG process closely mirrors key aspects of the workflow of an agent, as illustrated in Figure 1. For instance, the query rewriting phase in RAG, which involves semantic understanding and parsing, aligns with the semantic comprehension stage in the workflows of agents. Likewise, the retrieval phase of RAG, which focuses on extracting the most relevant documents, corresponds to the planning and execution phases of an agent, where decisions are made based on retrieved knowledge. Finally, the generation phase in RAG parallels the execution stage of an agent, where actions are performed based on the assigned task. This structural alignment suggests that the architecture of RAG is fundamentally converging with the workflows of agents, solidifying the position of RAG as a key facilitator of intelligent and autonomous systems. However, personalization within RAG and systems of agents presents fundamental complexities. First, these architectures require coordinated personalization across multiple interdependent stages, including pre-retrieval, retrieval, generation, planning, tool invocation, and memory management, where modifications at one stage cascade through subsequent operations. In practice, such personalization may arise either from incorporating a user-specific repository (such as long-term memory or personal notes) into the retrieval corpus or from conditioning query processing and retrieval or reranking over external knowledge using profiles, preferences, or interaction signals of users. This interdependency necessitates understanding how personalization propagates through the entire pipeline rather than optimizing isolated components. Second, in agentic systems, personalization extends beyond aligning generated content with user preferences to encompass the entire cycle

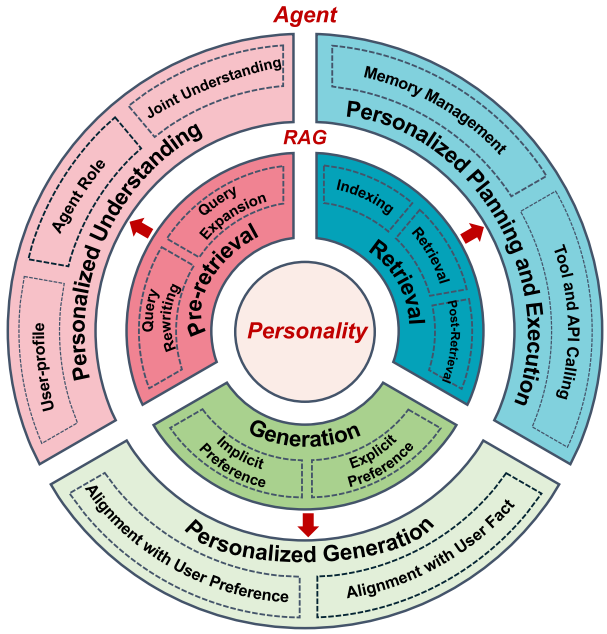

Fig. 1. Integration of personalization with RAG in the agent workflow. The personalization enhances three key RAG stages: *Pre-retrieval*, *Retrieval*, and *Generation*. This framework converges to a comprehensive agent workflow comprising *Personalized Understanding*, *Personalized Planning and Execution*, and *Personalized Generation*, which we denote as "Personalized RAG++."

of understanding, planning, execution, and generation. Agents must personalize not only what they communicate but also how they act, adapting decision-making strategies, tool selection, and task execution to align with individual intentions and expertise of users. These compounded challenges demonstrate that effective personalization in RAG and systems of agents demands holistic integration across all components rather than isolated interventions at individual stages.

However, there is currently a lack of a comprehensive survey that examines personalization from the perspectives of both RAG and agentic approaches. Existing surveys focus on different RAG methodologies [41, 45] or literature related to LLM-based agents [86, 168, 216], without systematically exploring their implications for personalization. Although recent surveys [80, 91, 217] discuss personalization, they primarily focus on utilizing LLM technologies for personalized generation through methods such as prompting, fine-tuning, and strategies for preference alignment. Furthermore, even when these surveys explicitly discuss RAG, they treat the framework as a single component within a broader landscape of personalization. They do not provide a workflow-centric, stage-by-stage analysis of where and how personalization is integrated across the complete RAG pipeline, nor do they detail how these mechanisms extend to agentic settings. Additionally, the work by Xu et al. [194] discusses personalized generation across different modalities but does not provide the comprehensive perspective required for this domain.

Motivated by the aforementioned gaps in the literature, this survey provides a comprehensive review of the integration of personalization within RAG and agentic RAG frameworks to enhance user experiences and optimize satisfaction. Our systematic collection of literature focuses on papers published after 2021 in top venues, including ICLR, SIGIR, NeurIPS, ACL, and TMLR. Given the rapid evolution of this field, we have also incorporated relevant preprints from arXiv published

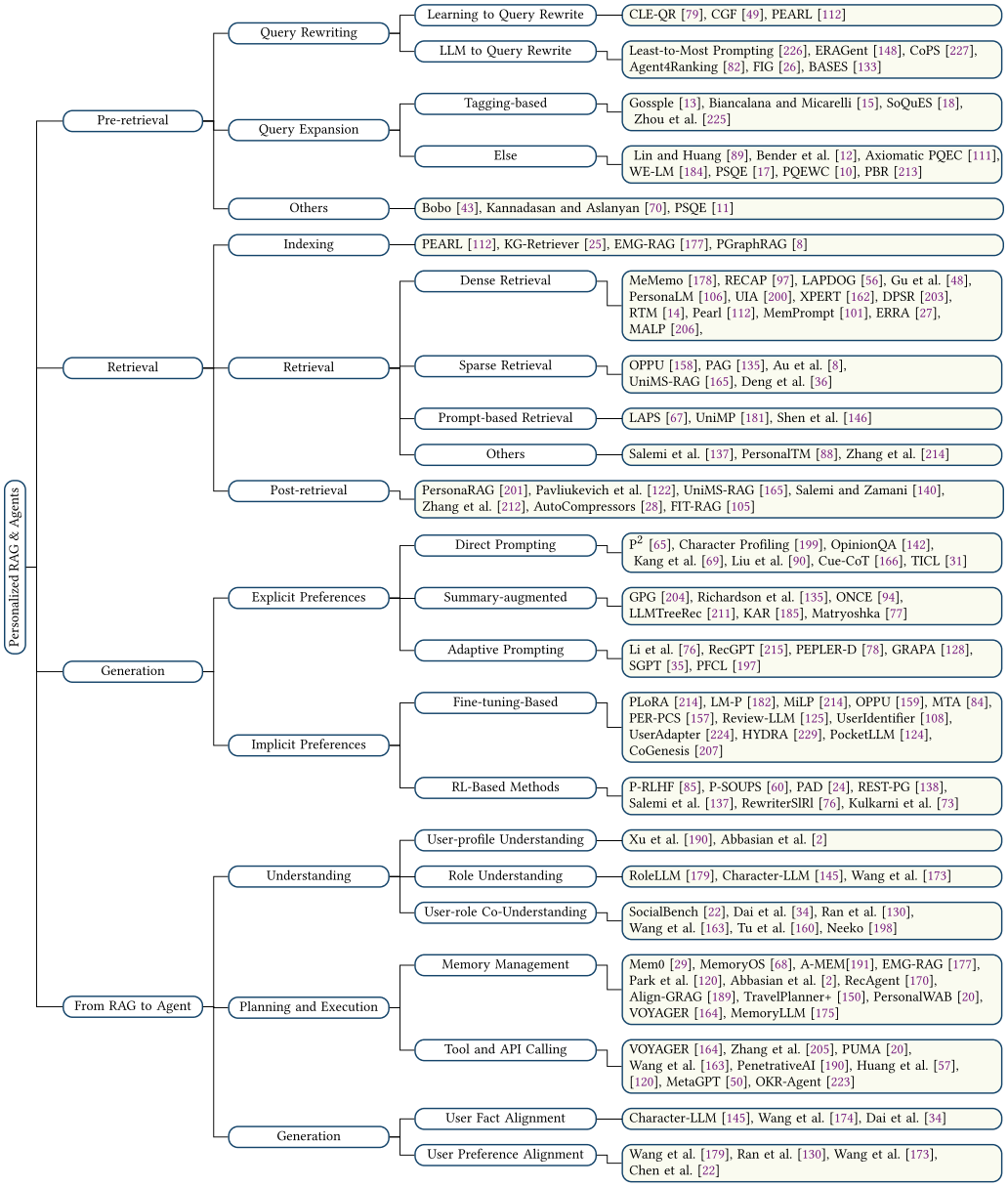

Fig. 2. A taxonomy of Personalized RAG and Agents with representative works.

after 2022. We apply strict selection criteria that require papers to demonstrate clear relevance to "Personalization" while intersecting with at least one of the following two domains: RAG systems or agent architectures. We analyze personalization through a structured examination of the RAG pipeline, following the processing sequence of information flow from input to output, and we identify the essential components of agent workflows where personalization mechanisms can be effectively integrated. The taxonomy and representative works are illustrated in Figure 2. This approach reveals how personalization can be systematically incorporated into existing frameworks

to create systems that are more adaptive and user-centric. The key contributions can be summarized as follows:

- We provide an extensive exploration of the existing literature on how personalization is integrated into various stages of RAG (pre-retrieval, retrieval, and generation) and agentic RAG (understanding, planning, execution, and generation).
- For each subtask, key datasets, benchmarks, and evaluation metrics are summarized, used in existing research to facilitate future studies in the respective domains.
- Current research limitations are examined alongside future directions for personalized RAG, focusing on potential advancements to address existing challenges.

The outline of this survey is as follows: we introduce what is personalization (Section 2) and explain how personalization is adopted into RAG pipeline (Section 3). Then, we present a literature review on where to integrate personalization within different stages of RAG and agentic RAG workflows (Section 4) and discuss the key datasets and evaluation metrics used in existing research (Section 5). Lastly, we present a discussion on the limitations of current research and future directions (Section 6).

2 What Is Personalization

Personalization refers to the adaptation of systems to the characteristics, preferences, and contexts of individual users, transforming generic solutions into tailored experiences. This paradigm has been fundamental in addressing information overload and enhancing user engagement across digital platforms [5, 123].

Personalized search and recommender systems have served as the primary domains for advancing personalization techniques. In personalized search, systems incorporate user profiles, search history, and behavioral patterns to rank or re-rank results beyond simple query-document matching [38]. Recommender systems have similarly evolved from collaborative filtering [143] and content-based approaches [123] to sophisticated deep learning models [113]. Notable examples include the recommender system based on deep neural networks from YouTube [32] and the recommendation framework of Spotify [107], which analyze temporal patterns, contextual features, and cross-domain preferences to model complex user–item interactions.

The importance of personalization is well-established through both research and commercial impact. Personalized recommendations drive up to 35% [100] of e-commerce revenue and significantly improve user retention on content platforms. Beyond business metrics, personalization addresses fundamental challenges in information discovery, task completion, and user satisfaction by filtering and prioritizing content according to individual relevance [5, 100].

The emergence of LLMs, RAG systems, and AI agents marks a paradigm shift in personalization capabilities. Unlike traditional systems that rank or filter existing content, these technologies can generate entirely new personalized content, engage in adaptive dialogue, and modify the reasoning processes for individual users [150, 182]. This evolution requires rethinking how personalization is conceptualized and implemented. In the context of RAG and agents, personalization extends beyond simple preference matching to encompass the dynamic incorporation of user-specific information throughout the processing pipeline. This involves tailoring both the retrieval mechanisms and the generation processes to align with the characteristics and needs of individual users.

To effectively implement such personalization, user information can be systematically categorized into the following types:

—Explicit User Profile: Explicitly presented information about users, including biographical details, attributes (e.g., age, location, gender, education), and social connections (e.g., social networks).
—User Historical Interactions: Behavioral data, including browsing history, clicks, and purchases, which help to infer the interests and preferences of users to improve personalization.
—User Historical Content: Implicit personalization derived from user-generated content, such as chat history, emails, reviews, and social media interactions.
—Persona-Based User Simulation: The use of agents based on LLMs to simulate and generate personalized interactions.

Integrating this personalized information at various stages of the workflows for RAG and agents enables dynamic alignment with human preferences, thereby making responses more user-centric and adaptive.

3 How to Adopt Personalization

We define the process of incorporating personalization into the RAG pipeline as follows:

$$g = \mathcal{G}\left(\mathcal{R}\left(\mathcal{Q}\left(q, p\right), C, p\right), \text{prompt}, p, \theta\right), \tag{1}$$

where p denotes the personalized information. This process unfolds in three steps. In the *pre-retrieval phase*, the query processing module ($\mathcal{Q}$) refines the initial query q using p, employing techniques such as query rewriting or expansion. During the *retrieval phase*, the retriever ($\mathcal{R}$) leverages p to obtain relevant documents from the corpus (C). Finally, in the *generation phase*, the retrieved information is combined with p and structured according to the provided prompt. This structured input is subsequently fed into the generator ($\mathcal{G}$), parameterized by θ, to produce the final response g. Consequently, personalized information exerts a direct influence on multiple stages of the RAG pipeline. In this survey, we consider an agent system as a specialized application of the RAG framework, wherein personalization is incorporated in a comparable manner.

4 Where to Adopt Personalization

4.1 Pre-Retrieval

4.1.1 Definition. Pre-retrieval constitutes a fundamental stage in information retrieval systems. In this phase, the original user query is refined or augmented prior to the retrieval process to enhance the relevance and quality of the search results, as illustrated in Figure 3. This procedure frequently incorporates additional contextual or personalized information to align the query more effectively with the intent of the user. The process is formally defined as follows:

$$q^* = \mathcal{Q}\left(q, p\right), \tag{2}$$

where p and q denote the personalized information and original query, and q^* is the optimized query after query reformulation.

4.1.2 Query Rewriting. Query rewriting within the pre-retrieval stage of RAG involves the reformulation of user queries to enhance retrieval effectiveness. This enhancement is achieved by improving relevance, disambiguating intent, or incorporating contextual information prior to the retrieval of documents from an external knowledge source. The existing literature regarding personalized query rewriting is broadly categorized into two primary classifications: (1) direct personalized query rewriting and (2) auxiliary personalized query rewriting.

(1) *Direct Personalized Query Rewriting.* The first category focuses on personalized query rewriting through the employment of direct models. For instance, Cho et al. [30] introduce a personalized search-based query rewriting system for conversational AI that addresses user-specific semantic

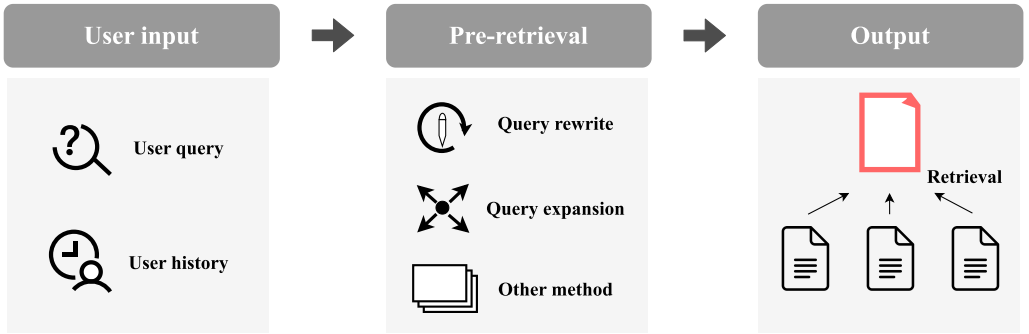

Fig. 3. Illustration of the personalized pre-retrieval stage. The system refines user queries by leveraging user history to perform *Query Rewriting* and *Query Expansion*, and other methods, enabling more accurate and relevant retrieval in the RAG framework.

and phonetic errors. Nguyen et al. [115] apply **Reinforcement Learning (RL)** techniques to improve query rewriting within online e-commerce systems, leveraging distilled LLM architectures for personalized performance. Similarly, CLE-QR [79] explores query rewriting within the Taobao search engine to enhance user satisfaction via customized query adaptation. CGF [49] introduces a constrained generation framework that facilitates more flexible and personalized query rewriting in the context of conversational AI. Furthermore, Li et al. [76] investigate learning methods to rewrite prompts for personalized text generation, thereby improving the relevance and engagement of AI-generated content. Additionally, PEARL [112] discusses the personalization of writing assistants based on LLMs through the integration of generation-calibrated retrievers, which enhances the quality of AI-generated content.

(2) *Auxiliary Personalized Query Rewriting.* The second category emphasizes personalized query rewriting through the utilization of auxiliary mechanisms, such as retrieval, reasoning strategies, and external memory. Zhou et al. [226] propose a least-to-most prompting strategy that assists in complex reasoning within LLMs, which is adaptable for personalized text generation. ERAGent [148] enhances retrieval-augmented LLMs to improve personalization, efficiency, and accuracy, thereby indirectly supporting personalized query rewriting for content generation. CoPS [227] integrates LLMs with memory mechanisms to create more personalized search experiences, which also influences content generation via improved query understanding. Moreover, Agent4Ranking [82] employs multi-agent LLM frameworks to perform semantic robust ranking, including personalized query rewriting to improve search rankings. FIG [26] combines graph-based methods with LLMs for query rewriting, improving personalized content generation and conversational interactions. Finally, BASES [133] employs LLM-based agents to simulate large-scale web search user interactions, contributing to the development of personalized query rewriting strategies for content generation.

4.1.3 Query Expansion. Query expansion enhances retrieval systems by augmenting the original query of a user with additional terms, synonyms, or a refined structure to capture intent more accurately. This procedure improves both the relevance and the scope of retrieved documents. Recent advancements in LLMs have reinvigorated this field [59, 62, 169], leveraging the comprehension and generation capabilities of these models to expand queries through encoded knowledge or external retrieval with notable success. Personalized query expansion, a specialized subset, incorporates user-specific data to tailor results, thereby boosting performance and customizing the search experience.

(1) *Tagging-Based Query Expansion.* By 2009, studies began incorporating tagging information to enhance personalized query expansion. For instance, Gossple [13] introduced the TagMap and TagRank algorithms, which dynamically select tags from personalized networks constructed via the cosine similarity of user–item tag distances, thereby improving recall performance. Similarly, Biancalana and Micarelli [15] recorded user queries and visited URLs, leveraging social bookmarking to extract relevant tags and construct a personalized three-dimensional co-occurrence matrix. Based on this structure, multiple semantically categorized expanded queries were generated to reflect the interests of the user more accurately. Further advancements include SoQuES [18], which integrated tag semantic similarity with social proximity, and a graph-based approach [225] that utilized Tag-Topic models and pseudo-relevance feedback for term weighting, tailoring the expansion process to individual user preferences.

(2) *Else.* Apart from tagging-based techniques, early research regarding personalized query expansion primarily focused on modeling user personalization based on search history [89], social networks, or preferences derived from friendship networks [12]. The Axiomatic PQEC framework [111] formalized expansion rules using both local (user behavior-driven) and social (network-driven) strategies. In 2017, WE-LM [184] advanced this paradigm by modeling multi-relational networks with word embeddings across tag-word relationships, refining associations through affinity graphs. Later, PSQE [17] further improved tagging-based methods using utf-iuf user profiling, integrating a tag similarity graph with user profiles in the online phase to compute expansion terms relevant to the interests of the user in real-time, achieving dynamic personalized expansion. In addition, PQEWC [10] leveraged clustering and contextual word embeddings to optimize query expansions dynamically. Recent studies, such as PBR [213], utilize personalized query expansion and incorporate RL-based optimization methods.

4.1.4 Others. Beyond query rewriting and query expansion, other personalized query-related research focuses on areas such as query disambiguation and query auto-completion [152]. Query disambiguation resolves ambiguous search queries by determining the actual intent of the user when multiple interpretations are possible. This approach leverages user context and preferences to select the most relevant meaning, although it may encounter difficulties with highly ambiguous queries or when insufficient user history is available. Query auto-completion predicts and suggests complete search queries in real-time as users type, personalizing suggestions based on search history and behavioral patterns; however, it may introduce bias toward popular queries and can be computationally expensive for real-time processing. Bobo [43] allows users to input contextual terms reflecting their domain knowledge for disambiguation purposes. In 2019, a method [70] applied fastText embeddings from recent queries to rank auto-completion candidates. In addition, PSQE [11] employed synthetic user profiles from Wikipedia and word2vec embeddings for query disambiguation.

4.1.5 Discussion. Query rewriting and query expansion represent two complementary paradigms within personalized pre-retrieval, each addressing distinct aspects of the semantic gap between the intent of the user and system understanding. Query rewriting fundamentally transforms the structure of a query to resolve ambiguity and correct misalignment. This approach is particularly valuable when users struggle to articulate their information needs or when contextual understanding is essential, such as in conversational search [52] and multi-turn interactions [102]. In contrast, query expansion preserves the core intent of the original query while enriching it with semantically related terms, thereby broadening the retrieval scope without fundamentally altering the expressed intent of the user. This distinction suggests that personalization strategies should be adaptive: employing rewriting for intent correction and disambiguation, while applying expansion

Table 1. Comparison of Personalized Pre-Retrieval Methods

Method	Description	Pros	Cons
		Query Rewriting	
Direct Query Rewriting	Reformulates using personalized methods	—Simple implementation —Handles semantic errors	—Needs extensive user data —Limited domain adaptation —Risk of overfitting
Auxiliary Query Rewriting	Uses external tools: retrieval, reasoning, memory	—Uses multiple sources —Better complex reasoning —More flexible and robust	—High computational cost —Increased latency
		Query Expansion	
Tagging-based Expansion	Expands using social tags and metadata	—Collective intelligence —Dynamic trend adaptation —Rich semantic relations	—Depends on tag quality —Vulnerable to noise
History/ Network-based Query Expansion	Uses user history, networks, embeddings or profiles	—Captures semantic similarity —Language-agnostic	—Compute intensive —Potential semantic drift —Hard to interpret
		Others	
Other Methods	Query disambiguation and auto-completion	—Improves user experience —Real-time guidance —Prevents early errors	—Limited scope —Privacy concerns —Constrains expression

for coverage and recall enhancement. The differences between these two paradigms are explicitly demonstrated in Table 1.

The evolution from early tagging-based approaches to techniques powered by LLMs reveals a critical insight: first, effective personalization requires not only the understanding of what users explicitly state but also the prediction of what they leave unexpressed. Modern systems increasingly benefit from hybrid approaches that dynamically select or combine both techniques based on query characteristics and user context. For instance, the recent work CHIQ [109] proposes a pipeline in which an ambiguous query first undergoes personalized rewriting to clarify the intent utilizing historical context, followed by expansion into multiple sub-queries to ensure comprehensive coverage. Second, beyond the selection of current technologies, there is a critical requirement to develop intelligent orchestration mechanisms capable of leveraging historical user data, contextual signals, and query properties to determine the optimal personalization strategy. Such mechanisms could potentially learn user-specific preferences [110, 213] regarding the degree of aggressiveness applied when rewriting versus expanding queries.

4.2 Retrieval

4.2.1 Definition. The retrieval process involves finding the most relevant documents D^* from a corpus C based on a query q^*, as shown in Figure 4. To incorporate personalization, the retrieval function $\mathcal{R}$ integrates additional user-specific information p. This allows the retrieval process to tailor the selected documents to align with individual user preferences or contexts, thereby enhancing the relevance and personalization of the generated outputs.

$$D^* = \mathcal{R}\left(q^*, C, p\right). \tag{3}$$

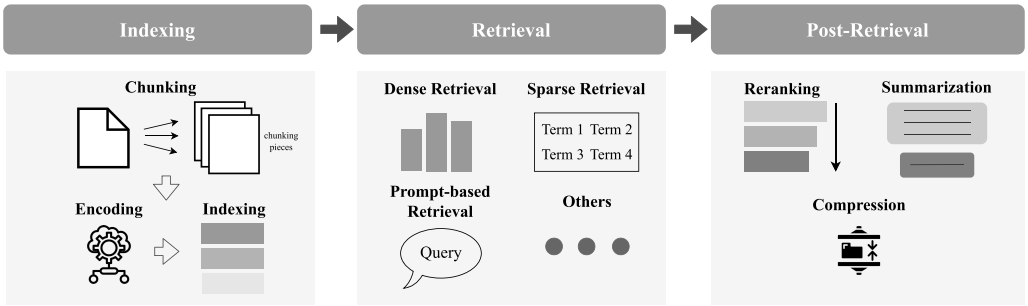

Fig. 4. Overview of the personalized retrieval stage. Including *Indexing*, which chunk or encode for document indexing. Then, different *Retrieval* methods (dense, sparse, prompt-based) are used and finally *Post-retrieval* processing for optimized information access.

In the retrieval process, personalization can primarily be introduced by focusing on three steps: indexing, retrieval, and post-retrieval. These steps ensure efficient and accurate retrieval of relevant documents or knowledge, while tailoring the process to individual user preferences. Below, we provide a detailed explanation of each step.

4.2.2 Indexing. Indexing organizes knowledge base data into a structured format to facilitate efficient retrieval. Within the RAG pipeline, documents are either chunked or entirely encoded into representations before being integrated into searchable systems [39, 154]. Conventional encoding methods employ either sparse encoding techniques (e.g., TF-IDF [129], BM25 [136]) or dense encoding approaches leveraging pre-trained models, such as BERT [1], Siamese Encoders [132], or LLM-based encoders [87, 187].

To introduce personalization at the indexing stage, PEARL [112] generates user embeddings by encoding personal history data with models like DeBERTa. These embeddings are subsequently clustered to create personalized shared indices. Other approaches integrate knowledge graphs into indexing to enhance retrieval performance. For example, KG-Retriever [25] employs a Hierarchical Index Graph, consisting of a knowledge graph layer and a collaborative document layer, to improve RAG retrieval. EMG-RAG [177] incorporates personalized memory within an editable knowledge graph, enabling dynamic retrieval. Similarly, PGraphRAG [8] leverages user-centric knowledge graphs to enhance personalization in retrieval tasks.

4.2.3 Retrieval. The Retrieval step matches a user query with the indexed knowledge base to fetch relevant candidates. It can be broadly categorized into four different types: (1) Dense Retrieval, (2) Sparse Retrieval, (3) Prompt-based Retrieval, and (4) Others.

(1) *Dense Retrieval.* Dense retrieval methods often use vector embeddings and similarity metrics (e.g., cosine similarity) and achieve personalization by encoding user preferences, context, or interactions into query or document embeddings, enabling tailored results through similarity-based matching. For instance, MeMemo [178] retrieves personalized information by matching user-specific embeddings with document vectors, focusing on private, on-device text generation. Similarly, RECAP [97] and LAPDOG [56] enhance personalized dialogue generation by encoding queries and user profiles as dense vectors and retrieving top-N results, ensuring user-specific context drives the responses. In chatbots, Gu et al. [48] integrates conversational context and user profiles to align retrieved responses with user personas. PersonaLM [106] employs group-wise contrastive learning, training its retrieval model to align user queries with domain-specific text fragments, thereby improving personalization. UIA [200] employs dual encoders to retrieve documents tailored to user

preferences. XPERT [162] incorporates temporal events and user interactions into embeddings, enabling large-scale retrieval across millions of items.

Dense retrieval also enhances specific applications like e-commerce, medical assistance, and language models. DPSR [203] and RTM [14] encode user queries and product information to personalize product searches dynamically. Pearl [112] and MemPrompt [101] retrieve personalized content by leveraging historical user data and memory-assisted mechanisms. ERRA [27] uses review embeddings as dense queries for recommendations. In medical assistance, MALP [206] integrates short and long-term user interactions into embeddings for contextualized, personalized responses.

(2) *Sparse Retrieval.* Sparse retrieval methods often rely on term-based matching (e.g., BM25) and apply personalization by assigning higher weights to terms or keywords that are more relevant to the user. OPPU [158] uses the BM25 algorithm to select the k most relevant records from the user's historical data for the current query. Similarly, PAG [135] incorporates user input and profiles to enhance summarization and retrieval, aligning sparse representations with personalization objectives for LLMs. Au et al. [8] uses BM25 search algorithms to find entries related to the target user or neighboring users through the graph structure. UniMS-RAG [165] combines sparse and dense retrieval by leveraging multi-source knowledge, such as dialogue context and user images, to refine personalized responses in dialogue systems. Lastly, Deng et al. [36] apply sparse retrieval to support fact-based queries, considering user queries and preferences to enhance answer generation for e-commerce applications.

(3) *Prompt-Based Retrieval.* Prompt-based retrieval leverages prompts to guide retrieval from the model or external sources and introduces personalization by crafting user-specific prompts that guide the retrieval process. These prompts may include explicit user preferences, historical interactions, or detailed instructions that reflect the user's unique requirements. By embedding this personalized context directly into the prompt, the retrieval process can dynamically adjust to capture and return results that are most relevant to the user. LAPS [67] focuses on multi-session conversational search by storing user preferences and dialogue context, then using prompts to retrieve relevant information tailored to the user's biases and categories of interest. UniMP [181] employs user interaction histories as input to prompt-based retrieval, enabling personalized recommendations for multi-modal tasks, such as vision-language applications, by aligning prompts with user behavioral data. In contrast, Shen et al. [146] explores the use of LLMs to extract empathy and narrative styles from user-provided stories, but this work primarily focuses on style extraction and does not explicitly involve a retrieval component.

(4) *Others.* RL-based retrieval personalizes the process by optimizing retrieval policies based on user feedback, learning user preferences over time to adjust strategies. Salemi et al. [137] combines models like BM25, RbR, and dense retrieval, refining them with RL and knowledge distillation to adapt to user profiles for personalized outputs. Parameter-based retrieval leverages pretrained model parameters to implicitly store and retrieve user-specific information, allowing direct retrieval from the model without traditional indices. PersonalTM [88] generates document identifiers (Document IDs) using a Transformer model, encoding query, history, and document relationships into its parameters for personalization. Similarly, Zhang et al. [214] uses parameterized representations to integrate user queries and histories, tailoring responses to individual preferences. Additionally, recent works such as SyNeg [81] explore the use of LLM-generated synthetic data to enhance the robustness and personalization of dense retrieval models.

4.2.4 Post-Retrieval. Current Post-retrieval methods primarily focus on refining retrieved documents or responses to improve relevance and coherence. Current methodologies could be categorized into three parts (1) Re-ranking, (2) Summarization, and (3) Compression.

(1) *Re-Ranking*. Re-ranking enhances personalized content generation by prioritizing more relevant documents at the top. PersonaRAG [201] extends RAG by integrating user-centric agents, such as the Live Session Agent and the Document Ranking Agent, to refine document ranking and improve overall performance. Pavliukevich et al. [122] propose a cross-encoder BERT model for re-ranking external knowledge within a personalized context. UniMS-RAG [165] introduces a scoring mechanism that evaluates retrieved documents and outputs by optimizing the retriever. Besides, it includes an evidence attention mask, enabling re-ranking during inference and applying it to personalized datasets. Salemi and Zamani [140] present an iterative approach to optimizing ranking results based on the expectation-maximization algorithm, with performance validated in personalized scenarios.

(2) *Summarization*. Summarization refers to the process of summarizing retrieved information to enhance performance. For instance, Zhang et al. [212] introduced a role-playing agent system to summarize retrieved history in order to improve the final Personalized Opinion Summarization process.

(3) *Compression*. Compression involves condensing embeddings or retrieved content to enhance efficiency and effectiveness. Approaches like AutoCompressor [28] compress contextual embeddings into shorter semantic representations, and FIT-RAG [105] introduces a self-knowledge recognizer along with a sub-document-level token reduction mechanism to minimize tokens within RAG pipeline. However, few studies have specifically explored personalized fields, highlighting a promising direction for future research.

4.2.5 Discussion. Indexing, retrieval, and post-retrieval methods each play a critical role in ensuring efficient and personalized information processing, with specific applications and tradeoffs. Indexing focuses on organizing knowledge bases for efficient retrieval, using techniques such as sparse encoding methods like TF-IDF and BM25, which are efficient but limited in understanding semantics, and dense encoding methods like BERT and DeBERTa, which provide better semantic understanding but require significant computational resources. These methods are widely used in tasks like question answering and personalized recommendation systems. Retrieval involves matching user queries with relevant documents and can be categorized into dense retrieval, which provides high semantic understanding and personalization but is computationally expensive; sparse retrieval, which is efficient and interpretable but less capable of handling semantics; prompt-based retrieval, which is highly flexible and adaptable to user needs but requires careful engineering of prompts; and advanced methods like RL-based approaches, which dynamically adapt to user feedback but are complex to implement. This step is essential in applications like personalized dialogue systems, search engines, and e-commerce. Post-retrieval methods refine retrieved results to enhance relevance and coherence through re-ranking, which improves personalization and prioritizes relevant content but increases computational overhead; summarization, which simplifies complex information for better user understanding but risks losing critical details; and compression, which reduces computational costs by condensing information but remains underexplored in personalized contexts. Together, these methods provide a comprehensive pipeline for delivering efficient, relevant, and personalized outputs, balancing their strengths in semantic understanding, relevance, and flexibility with challenges related to computational costs and implementation complexity. We clearly demonstrate method differences in Table 2.

Recent literature reveals trajectories toward hybrid and adaptive architectures that transcend the limitations of individual approaches. Firstly, the integration of knowledge graphs into indexing systems, like KG-Retriever [25], EMG-RAG [177], and PGraphRAG [8], represents a significant advancement beyond traditional vector-based representations, enabling more structured and interpretable personalization. Secondly, there is an observable convergence of sparse and dense

Table 2. Comparison of Personalized Retrieval Methods

Method	Description	Pros	Cons
		Indexing	
Sparse Encoding	Term-based methods (TF-IDF, BM25)	—Efficient computation —Highly interpretable —Fast indexing	—Limited semantics —Poor with synonyms —Vocab mismatch
Dense Encoding	Embedding-based (BERT, DeBERTa)	—Strong semantics —Handles synonyms —Better personalization	—High compute cost —Large storage needs —Slower indexing
		Retrieval	
Dense Retrieval	Vector similarity matching	—Excellent matching —Highly personalizable	—Computationally expensive —Needs quality embeddings —Black-box nature
Sparse Retrieval	Keyword-based matching (BM25)	—Fast and scalable —Transparent results —Low resource usage	—Weak semantics —Exact match dependency —Limited personalization
Prompt-based	Uses prompts to guide retrieval	—Highly flexible —Dynamic adaptation	—Prompt engineering needed —Inconsistent results —Context length limits
		Post-retrieval	
Re-ranking	Reorders retrieved documents	—Improves relevance —Personalized ordering —Corrects initial errors	—Additional latency —Compute overhead —Complex optimization
Summarization	Condenses retrieved information	—Better understanding —Reduces info overload —Focuses on key points	—Risk of info loss —Quality varies —Context dependency
Compression	Reduces embeddings/ content size	—Lower compute cost —Faster processing	—Potential quality loss —Underexplored area

retrieval paradigms, with work like UniMS-RAG [165] demonstrating that multi-source knowledge integration can leverage the computational efficiency of sparse methods while maintaining the semantic richness of dense approaches. Thirdly, the recent emergence of RL-based retrieval systems signals a fundamental shift from static retrieval policies to dynamic and adaptive strategies that continuously refine themselves based on feedback [55, 66]. This evolution suggests that future personalized retrieval systems will increasingly prioritize end-to-end optimization rather than treating indexing, retrieval, and post-retrieval as isolated components.

4.3 Generation

4.3.1 Definition. Personalized generation incorporates user-specific retrieved documents D^*, task-specific prompt, and user preference information p via the generator $\mathcal{G}$ parameterized by θ to produce tailored content g^* aligned with individual preference, where the flow is shown in

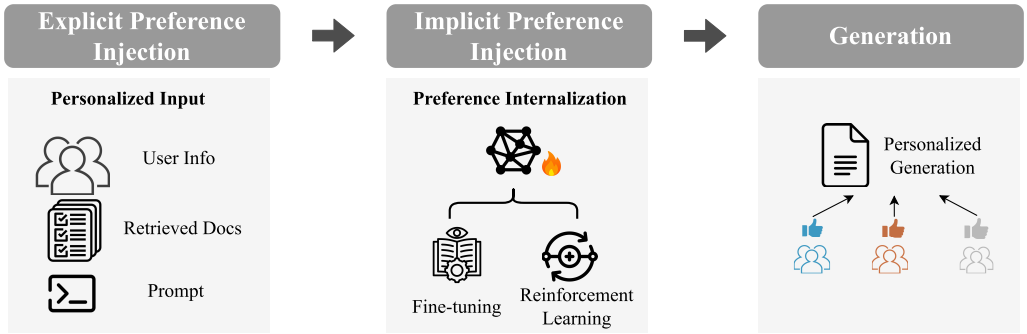

Fig. 5. Overview of the personalized generation stage. *Explicit Preferences Injection* denotes preference is injected through personalized inputs (user info, retrieved documents, prompts), while *Implicit Preferences Injection* denotes preference is internalized via fine-tuning and RL, leading to personalized content generation.

Figure 5. The generation process can be formulated as:

$$g^* = \mathcal{G}\left(D^*, \text{prompt}, p, \theta\right). \tag{4}$$

Personalized generation can be achieved by incorporating explicit and implicit preferences. Explicit preference-driven methodologies utilize direct input signals (e.g., D^*, prompt, and p), to tailor outputs to specific user preferences. Conversely, implicit preference-encoded approaches embed personalized information within the parameters θ of the generator model, during training, thereby facilitating preference alignment without the necessity for explicit runtime inputs.

4.3.2 Generation from Explicit Preferences. Integrating explicit preferences into LLMs facilitates personalized content generation. Explicit preference information encompasses demographic data of users (e.g., age, occupation, gender, and location), historical behavioral sequences (which reflect past behavioral patterns), and historical text outputs (which capture writing styles and tonal preferences). The integration of explicit preferences for personalized generation can be categorized into three approaches: (1) Direct-integrated Prompting, (2) Summary-augmented Prompting, and (3) Adaptive Prompting.

(1) *Direct-Integrated Prompting.* Integrating explicit preferences of users into language models through prompting enables the prediction of intents and behavioral patterns, facilitating personalized content generation. For instance, P^2 [65], Character Profiling [199], and OpinionQA [142] integrate personalized data into LLMs through prompting for role-playing tasks, thereby aligning the responses of the model with specified profiles of users. Kang et al. [69, 90] integrate interaction histories into LLMs via prompting to predict ratings for candidate items. Cue-CoT [166] employs chain-of-thought reasoning to infer needs from contextual cues, enabling personalized responses to in-depth dialogue questions. Additionally, TICL [31] proposes a trial-and-error framework that critiques initial responses generated by LLMs, derives explanations, and integrates these negative examples into prompts to improve personalization alignment.

(2) *Summary-Augmented Prompting.* The direct integration of personalized information via prompting struggles with ambiguous intent signals: lengthy interaction histories introduce noise that obscures critical behavioral patterns [93], while sparse behavioral data lacks sufficient context for LLMs to derive meaningful preferences of users. To address these issues, recent approaches focus on summarizing the personalized intents of users and integrating them into prompts. For instance, GPG [204] extracts key habits and preferences from personal contexts, enabling fine-grained personalization. Similarly, LLMs are employed to generate task-specific summaries of

user preferences, enhancing retrieval-augmented capabilities for personalized generation [135]. In recommendation systems, ONCE [94], LLMTreeRec [211], and KAR [185] leverage historical interactions between users and items to summarize preferences. Furthermore, Matryoshka [77] generates summaries of preferences by dynamically retrieving and synthesizing historical data.

(3) *Adaptive Prompting*. The manual design of personalized prompts demands both expert knowledge and significant labor, motivating the development of automated methods for the generation of personalized prompts. For example, [76] trains a personalized prompt rewriter via supervised and RL. RecGPT [215] and PEPLER-D [78] leverage prompt tuning to generate personalized prompts, enhancing sequential and explainable recommendations, respectively. GRAPA [128] integrates semantic and collaborative signals from interaction graphs between users and items with graph neural networks to generate context-aware personalized prompts. SGPT [35] employs prompt tuning to jointly model common and group-specific patterns, bridging generalized and personalized paradigms of federated learning. Furthermore, PFCL [197] achieves the modeling of human preferences across multiple granularities: coarse-grained prompts distill shared knowledge, while fine-grained prompts adapt to the characteristics of individual users.

4.3.3 Generation from Implicit Preferences. Unlike explicit preference modeling, which captures user preferences through textual input, implicit preference-based methods incorporate personalization through internal model parameters. In the context of personalized RAG, two primary approaches have emerged. The first approach focuses exclusively on parameter-side optimization and has been widely adopted in Personalized LLMs, with substantial research contributions in this area. The second approach involves collaborative optimization of both parameter-side and input-side components, where retrieval information is incorporated during parameter optimization to achieve enhanced personalization. This latter approach has received comparatively less research attention. In this work, we adopt two parameter-side optimization strategies to introduce implicit preference injection methods for these two categories of work. Specifically, we introduce finetuning-based methods (e.g., LoRA [53]) and preference alignment methods [24, 76] to examine how personalized information can be integrated into both implicit preference modeling frameworks.

(1) *Fine-Tuning Based Methods*. Most fine-tuning methods for personalization focus on parameter-side optimization to enhance model performance. Notable examples include PLoRA [214], which introduces a personalized knowledge integration framework combining task-specific LoRA with user-specific knowledge, and LM-P [182], which personalizes information via LoRA by incorporating User ID as a personalization factor. PER-PCS [157] extends this approach by incorporating a gating module that selects the appropriate LoRA adapter, enabling more fine-grained personalization, and MTA adopts a LoRA merging step to enhance more flexible personalization [84]. Beyond LoRA-based approaches, alternative architectures have emerged for personalized generation. UserIdentifier [108] introduces user-specific identifiers that significantly reduce training costs while enhancing personalized demonstrations. UserAdapter [224] proposes user-independent prefix embeddings that leverage prefix tuning for personalization, while HYDRA [229] and PPlug [92] achieve implicit personalization through training user-specific headers and embeddings. Recent research has further explored fine-tuning personalized models on edge devices [124] and collaborative learning between small and LLMs [207] to enable more effective personalized generation.

However, some other works have adopted hybrid approaches that combine both input-side retrieval and parameter-side optimization. These methods leverage retrieved personalized information as supplementary context while simultaneously optimizing model parameters. For instance, MiLP [214] employs Bayesian optimization to determine the optimal personalization injection configuration, including LoRA settings, to effectively capture and utilize retrieved user-specific information. Similarly, OPPU [159] combines non-parametric user knowledge retrieval with parameter-side

optimization to boost overall performance. These hybrid approaches demonstrate that integrating retrieval mechanisms with parameter fine-tuning can offer complementary benefits for personalized language model generation.

(2) *RL Based Methods.* Apart from fine-tuning based methods, recent research has explored RL based techniques on parameter side to personalize text generation by aligning outputs with user preferences. P-RLHF [85] has been proposed to jointly learn a user-specific and reward model to enable text generation that aligns with a user's styles or criteria. P-SOUPS [60] models multiple user preferences as a Multi-Objective RL problem, decomposing preferences into multiple dimensions, each trained independently. PAD [24] aligns text generation with human preferences during inference by utilizing token-level personalized rewards to guide the decoding process. REST-PG [138] introduces a framework that trains LLMs to reason over personal data during response generation. This approach first generates reasoning paths to enhance the LLM's reasoning ability and then employs Expectation-Maximization Reinforced Self-Training to iteratively refine the model based on its high-reward outputs. Additionally, Salemi et al. [137] incorporates RL into the RAG pipeline to improve retrieval accuracy, thereby enhancing the personalization of generated content. Other applications include RewriterSlRl [76], which has been introduced to generate text via RL-based personalized prompt rewriting using API-based LLMs, and Kulkarni et al. [73], who explore the use of RL to optimize RAG for improving the relevance and coherence of chatbot responses in specialized domains, ultimately enhancing user satisfaction and engagement.

4.3.4 Discussion. Personalized generation can be adopted via both explicit and implicit preference injection, yet they exhibit distinct characteristics that make them suitable for different scenarios. In explicit preference-based generation, personalization is clearly defined through user profile descriptions, contextual information, and similar inputs, which are incorporated into generators via prompts. A key advantage of this approach is explainability, as the personalized information is explicitly provided and easily traceable. Despite leveraging provided preferences and internal knowledge, explicit preference injection's personalization is constrained by model capabilities and irrelevant information interference. In contrast, implicit preference-based generation internalizes personalized information into the generator's parameters through scene-specific personalized data, thereby adapting the model for more fine-grained personalization. However, these methods typically incur substantial training and computational costs, as they require fine-tuning the generator's internal parameters. Therefore, selecting between these approaches should be guided by the specific application scenario and resource constraints. We clearly demonstrate method differences in Table 3.

A notable trend in recent literature is the convergence toward hybrid architectures that strategically combine explicit and implicit personalization mechanisms. Recent surveys and comparative studies [141, 153, 217] have converged on a practical division of labor: relatively stable, long-horizon preferences are compactly encoded using PEFT such as LoRA, while rapidly shifting, context-dependent signals are injected through prompting and retrieval at inference time. Recent hybrid memory research [218] further demonstrates that pairing lightweight fine-tuning with selective retrieval substantially improves fidelity on multi-hop and user-specific reasoning tasks, offering a practical pathway to achieve fine-grained yet agile personalization. However, they all face a fundamental challenge that parameter-level encoding yields deeper alignment with individual preferences but introduces higher computational and infrastructure costs while reducing transparency, whereas explicit prompting and RAG retain explainability and deployment flexibility at the expense of context length constraints and retrieval noise sensitivity. To reconcile these tradeoffs, recent works also explore preference optimization techniques such as online DPO [21], which enable

Table 3. Comparison of Personalized Generation Methods

Method	Description	Pros	Cons
		Explicit Preference-Based Generation	
Direct-integrated Prompting	Integrates user data directly into prompts	—High explainability —No training required —Easy to implement	—Context length limits —Noise from long history —Ambiguous intent signals
Summary-augmented Prompting	Summarizes preferences before integration	—Reduces noise —Better intent extraction —Handles long histories	—Info loss in summary —Extra processing overhead —Depends on summary quality
Adaptive Prompting	Automatically generates personalized prompts	—No manual design needed —Multi-granularity modeling —Context-aware adaptation	—Requires training —Higher compute cost
		Implicit Preference-Based Generation	
Fine-tuning Based	Uses fine-tuning methods like PEFT	—Fine-grained personalizing —Parameter efficient —Persistent preferences	—Training costs —Less explainable —Model-specific adaptation
Reinforcement Learning Based	Aligns generation with preferences via RL	—Dynamic preference learning —Multi-objective optimization —Continuous improvement	—High compute demands —Training instability —Complex reward design

systems to track preference drift and update policies incrementally without costly full retraining cycles.

4.4 From RAG to Agent

4.4.1 Definition. A personalized LLM-based agent is a system designed to dynamically incorporate user context, memory, and external tools or APIs to support highly personalized and goal-oriented interactions [23, 58, 186], and solve problems in a goal-oriented manner [86, 149]. To clarify the scope of this survey, we define agents based on three distinguishing characteristics: goal-directed behavior beyond single-turn responses, persistent memory across interactions, and the capacity to plan actions or invoke external tools. This encompasses a spectrum from conversational dialog agents that maintain user context, to role-playing agents requiring consistent persona maintenance, to task-oriented planning agents that decompose complex goals, to tool-augmented agents that orchestrate external APIs. Personalization manifests differently across this spectrum: conversational agents require linguistic style adaptation, planning agents need constraint satisfaction aligned with user goals, and tool-augmented systems must filter resources based on user history. The three-stage framework we present applies across these agent types but with varying emphasis depending on their primary function and autonomy level. From the previously introduced stages of RAG, we observe that the evolution of personalized RAG reveals a structural convergence with agent architectures. We analyze them from three key perspectives:

— *Personalized Understanding*: This phase within the agent parallels the query understanding and rewriting process of RAG as outlined in Section 4.1. However, it extends beyond static semantic parsing by incorporating dynamic user profiling [179] and role modeling [145]. This integration enables the agent to dynamically align interactions with implicit user preferences, facilitating personalized responses and task-specific adaptations [130].

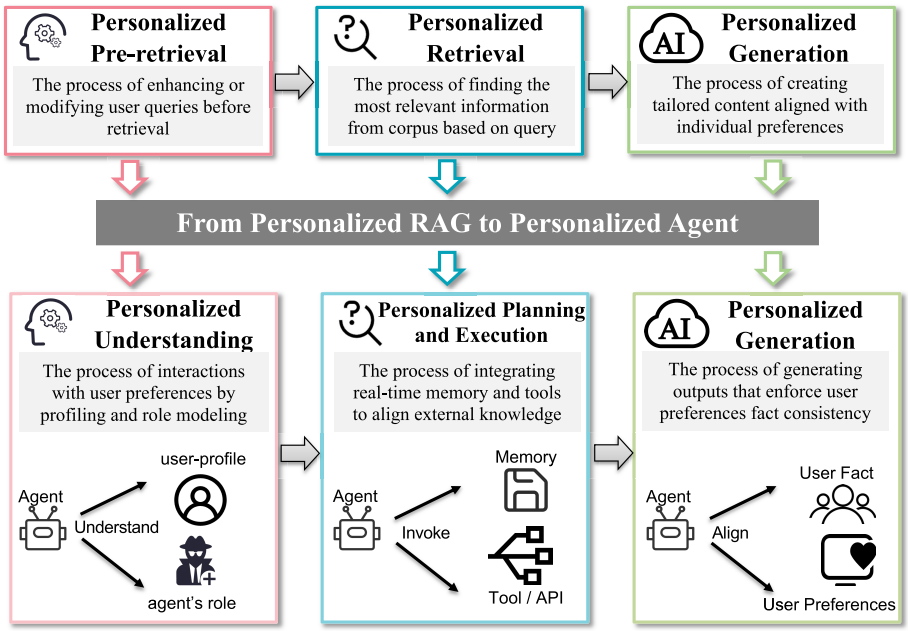

Fig. 6. Overview of the transition from personalized RAG to a personalized agent. The framework evolves from personalized Pre-retrieval, Retrieval, and Generation toward an agent capable of *Personalized Understanding*, *Planning and Execution*, and *Generation* through user profiling, memory integration, and preference alignment.

— *Personalized Planning and Execution*: This phase in agents mirrors RAG's retrieval process in Section 4.2 yet it advances beyond static document retrieval by incorporating real-time memory management [120] and sophisticated tool and API calling [164]. This approach ensures the dynamic alignment of external knowledge with personalized constraints, such as integrating medical history in healthcare agents [2], to deliver context-aware and user-specific outcomes.
— *Personalized Generation*: This phase in agents mirrors RAG's generative process in Section 4.3 but transcends static template-based generation by integrating user preference and fact alignment. Agents dynamically enforce user preferences and ensure fact consistency through role-specific mechanisms (e.g., social adaptability in conversational agents [2]), enabling outputs to evolve in harmony with personalized and situational constraints rather than relying solely on predefined generative frameworks.

In general we frame agent architectures as "*personalized RAG++*," where persistent memory [177] replaces static indexes, and tool APIs [20] serve as dynamic knowledge connectors, enabling complicated, human-aligned interactions beyond one-shot retrieval, as shown in Figure 6. This progression highlights that as RAG systems incorporate deeper personalization—requiring user-state tracking, adaptive tool usage, and context-aware generation, they inherently adopt agent-like capabilities.

4.4.2 Personalized Understanding. Personalized understanding refers to an agent's ability to accurately interpret user inputs by integrating user intent recognition and contextual analysis. This process ensures interactions that are both meaningful and contextually appropriate. The rationale behind this classification lies in its capacity to address three core aspects of understanding:

recognizing user intent, analyzing context, and leveraging user profiles. Each of these aspects plays a distinct role in improving the agent's performance.

(1) *User-Profile Understanding.* In user-profile understanding, an agent's personalized ability primarily depends on its capacity to accurately model and understand the user's preferences, context, and intentions. Xu et al. [190] proposes a framework in which LLMs are designed to understand the physical world, thereby facilitating a deeper connection between the agent and its environment, which is essential for accurate task execution. Abbasian et al. [2] further expands this understanding by emphasizing the importance of personalization in health agents, where the user's profile directly influences the behavior and decisions of the agent. This user understanding is foundational to ensuring that the AI agent performs tasks in a way that aligns with individual user needs.

(2) *Role Understanding.* In agent's role understanding, the role of the agent within these environments is also crucial. Recent studies focus on enhancing role-playing capabilities within LLMs. Wang et al. [179] introduces RoleLLM, a benchmark that aims to elicit and refine the role-playing abilities of LLMs, demonstrating how role understanding influences agent performance in conversational tasks. Similarly, Shao et al. [145] presents Character-LLM, a trainable agent framework for role-playing, which tailors its responses based on predefined roles. Wang et al. [173] introduces a method for evaluating personality fidelity in role-playing agents through psychological interviews, aiming to enhance the realism and consistency of AI-driven characters. This role understanding allows for more contextually appropriate interactions, increasing the relevance and utility of AI agents across various applications.

(3) *User-Role Joint Understanding.* In agent's user-role joint understanding, the intersection of user and role understanding is explored through frameworks that evaluate and enhance the social and personality aspects of LLMs. SocialBench, Chen et al. [22] provides a sociality evaluation framework for role-playing agents. Dai et al. [34, 130] extend this by incorporating multi-modal data and personality-indicative information, respectively, which allows agents to better adapt to both user and role understanding in dynamic environments. Furthermore, Wang et al. [163] offers a perspective on how role and environment understanding can improve user experience. Tu et al. [160] contributes by providing a benchmark specifically for evaluating role-playing agents in the Chinese context, adding a cultural dimension to role understanding. Finally, Neeko [198] further advances role-based interactions.

4.4.3 Personalized Planning and Execution. Personalized planning and execution refer to the process of designing and implementing strategies or actions that are specifically tailored to an individual's unique context, and goals [57, 120, 150, 205]. It requires agents to dynamically integrate long-term memory, real-time reasoning, and external tool utilization [50, 51, 223], as demonstrated in healthcare decision support [2] and travel planning scenarios [20]. We analyze two fundamental components that enable this personalization in the following.

(1) *Memory Management.* Effective memory systems allow agents to integrate users' historical preferences, behavioral patterns, and contextual habits, enhancing their ability to make planning and tailor interactions to user-specific needs [20, 164, 175]. Recent representative agent-memory frameworks have begun to treat long-term memory as a first-class, system-level component rather than an *ad-hoc* retrieval buffer. Mem0 [29] introduces a scalable, memory-centric pipeline that extracts and consolidates user interaction facts for on-demand retrieval, featuring a graph-based variant to capture relational structures. Following this paradigm, Align-GRAG [189] builds upon these foundations to further align graph-based knowledge with RAG. In parallel, MemoryOS [68] frames memory as an "operating system" problem and proposes comprehensive memory management with a hierarchical organization to support long-term conversational coherence and persona

persistence. A-MEM [191] explores agentic memory organization, where memories are written as structured notes, linked into a growing network, and can be updated as new experiences arrive. The EMG-RAG framework [177] combines editable memory graphs with RAG to maintain dynamic user profiles, while Park et al. [120] implements memory streams and periodic reflection mechanisms to simulate human-like behavior. In healthcare applications, Abbasian et al. [2] integrates multimodal user data through specialized memory modules to optimize treatment recommendations. For recommendation systems, RecAgent [170] employs hierarchical memory structures to model user interaction patterns across multiple domains. Recent advances like TravelPlanner+ [150] demonstrate how memory-augmented LLMs achieve higher relevance in personalized itinerary generation compared to generic planners.

(2) *Tool and API Calling*. The integration of external tools expands agents' capabilities beyond pure linguistic reasoning, enabling agents to interact with users and perform personalized tasks [20, 163, 164, 190, 205]. For instance, VOYAGER [164] establishes a paradigm for lifelong skill acquisition through automatic API curriculum learning and skill library construction. In robotics, Zhang et al. [205] develop a bootstrapping framework where LLMs guide robots in tool-mediated skill discovery, enabling a high success rate in novel object manipulation tasks. The PUMA framework [20] demonstrates how personalized web agents can achieve performance gains in e-commerce tasks through adaptive API orchestration. For mobile interaction, Wang et al. [163] implements few-shot tool learning to handle diverse UI operations with minimal training data. These approaches highlight the importance of tool grounding mechanisms [57] that translate linguistic plans into executable API sequences while maintaining personalization constraints.

This synthesis highlights that modern agent systems achieve enhanced personalization through two primary strategies: (1) Memory-augmented architectures, which leverage editable memory graphs [177], reflection mechanisms [120], and hierarchical memory structures [170] to dynamically adapt to user preferences across various domains; and (2) Tool and API integration, which expand agent capabilities by balancing generalization with specialization. Future work may explore improving the contextual relevance and adaptability of memory systems while optimizing real-time tool interaction for seamless task execution.

4.4.4 Personalized Generation. Based on the foundation of personalized planning and execution mechanisms, which enable agents to adapt strategies to user-specific contexts [57, 205], the next critical concern lies in personalized generation. This capability ensures that generated outputs not only align with factual correctness but also resonate with users' unique preferences, personality traits, and situational needs. Personalized generation bridges the gap between adaptive reasoning and human-aligned outcomes, allowing agents to produce contextually relevant and emotionally appropriate responses.

(1) *Alignment with User Fact*. Alignment with User Fact emphasizes the accuracy, consistency, and factual grounding of personalized responses, ensuring they remain trustworthy across diverse user interactions. This is particularly challenging in personalized agents, where maintaining character authenticity while avoiding hallucinations requires balancing creativity with factual adherence. Recent advances address these challenges through improved training frameworks and evaluation metrics. For instance, Character-LLM [145] integrates memory-augmented architectures to reduce hallucinations while preserving character-specific traits. Wang et al. [174] investigates quantization effects on personality consistency in edge-deployed agents and stabilizes outputs under computational constraints. Dai et al. [34] ensures multimodal consistency (text-image) in role-playing. These works highlight the importance of architectural innovations and rigorous evaluation in achieving reliability.

Table 4. Comparison of Personalized Agent Methods

Method	Description	Pros	Cons
		Personalized Understanding	
User-profile Understanding	Models preferences and context for task alignment	—Direct preference capture —Individual need alignment —Context-aware responses	—Limited real-time adaptation —Privacy concerns —Data dependency
Role Understanding	Enhances role-playing capabilities	—Contextual interactions —Consistent behavior —Enhanced realism	—Static role definitions —Limited generalization —Training complexity
User-role Joint Understanding	Combines user and role aspects	—Multi-modal adaptation —Cultural awareness —Social adaptability	—Regional specific limits —Complex evaluation —Higher compute needs
		Personalized Planning and Execution	
Memory Management	Integrates historical preferences and patterns	—Dynamic user profiling —Long-term context retention —Behavior pattern learning	—Memory scalability issues —Update latency —Storage overhead
Tool and API Calling	Enables external tool integration	—Extended capabilities —Real-world interaction —Task automation	—API dependency —Cold-start problem —Limited emergent discovery
		Personalized Generation	
Alignment with User Fact	Ensures factual accuracy and consistency	—Reduced hallucinations —Trustworthy outputs —Character authenticity	—Creativity constraints —Compute overhead —Balance complexity
Alignment with User Preferences	Reflects individual values and styles	—Personalized interactions —Dynamic adaptation —Enhanced engagement	—Risk of bias reinforcement —Ethical boundary detection —Over-fitting to preferences

(2) *Alignment with User Preferences.* Alignment with user preferences ensures that generated outputs reflect individualized personalities, values, and interaction styles. This requires agents to dynamically interpret implicit user cues and adapt responses accordingly. Wang et al. [179] benchmarks role-specific alignment. Ran et al. [130] improve personality fidelity via psychological scale datasets. Wang et al. [173] quantify alignment via psychological interviews. Chen et al. [22] evaluate social adaptability in conversations.

4.4.5 Discussion. The architectural evolution from RAG to personalized agents introduces significant advancements in human-AI interaction but also surfaces critical challenges that warrant further investigation. We provide a comparison of different strategies in Table 4 for reference.

Personalized Understanding, while enabling interpretation of user intent and context, faces limitations in real-time adaptability and generalization. Current approaches like RoleLLM [179] and Character-LLM [145] demonstrate robust role-specific comprehension but struggle with temporal dynamic user state tracking, particularly when handling evolving preferences or multi-session interactions. Furthermore, cultural specificity in benchmarks like CharacterEval [160] reveals gaps in global applicability, as agents trained on region-specific data often fail to generalize across diverse sociocultural contexts. This suggests that future architectures must move beyond simple fine-tuning and instead dynamically integrate cultural knowledge through cross-cultural transfer learning strategies that preserve personalization while enabling global scalability. Future work

could explore hybrid architectures that combine continuous learning mechanisms with privacy-preserving federated learning to address these adaptability constraints while maintaining user trust.

Personalized Planning and Execution, achieves remarkable task specialization through memory management and tool integration, yet suffers from scalability issues in complex environments. While frameworks like EMG-RAG [177] and VOYAGER [164] effectively manage user-specific constraints, their reliance on predefined API taxonomies limits emergent tool discovery in novel scenarios. The "cold-start" problem persists in domains requiring rapid skill acquisition, as seen in healthcare applications [2], where delayed API responses can compromise decision-making efficacy. A promising direction involves developing meta-reasoning architectures that dynamically prioritize memory recall versus tool invocation based on situational urgency and confidence thresholds.

Personalized Generation balances factual accuracy with preference alignment but risks over-fitting [91], where excessive finetuning creates self-reinforcing feedback loops that amplify existing biases while diminishing exposure to diverse perspectives. Techniques address surface-level alignment but lack mechanisms for ethical boundary detection. For instance, agents might inadvertently propagate harmful stereotypes when mirroring user preferences without critical oversight. Future systems could integrate value-aligned RL with human-in-the-loop validation to preserve authenticity while preventing detrimental customization.

5 Data and Evaluation Methodology

In the evolving landscape of personalization, spanning from RAG to advanced agent-based systems, model evaluation depends critically on diverse datasets and task-specific metrics. In our survey, we provide a comprehensive collection of these evaluation resources, organized by usage stage and offering essential insights for personalized tasks. In the following sections, we examine each stage of RAG and agentic systems in detail, presenting relevant datasets, metrics, and corresponding analyses.

5.1 Pre-Retrieval

Table 5 presents the comparison of datasets. Regarding methods of pre-retrieval, the approach to evaluation differs fundamentally between query rewriting and query expansion. Research on rewriting typically begins by employing metrics of text similarity as proxies for quality. However, to demonstrate genuine utility, these studies must ultimately validate findings against the downstream performance of retrieval. In contrast, research on expansion directly targets metrics of retrieval from the outset. Nevertheless, this approach faces the critical challenge of preventing semantic drift, a phenomenon where topically related terms fail to align with the specific needs for personal information. Regarding the selection of datasets, we observe a clear progression across three tiers of personalization intensity. At the foundational level, synthetic or weakly personalized environments, such as SCAN [75, 226] and Robust04 [82], rely primarily on metrics of accuracy. The intermediate tier encompasses domains with moderate signals, including Amazon Review [76, 116], Reddit Comments [76, 155], and ESCI [115, 131]. Within these domains, metrics for text generation, such as BLEU and ROUGE, serve to evaluate the fidelity of rewriting. At the most sophisticated level, strongly personalized environments, represented by AOL Query Log [121, 227], WARRIORS [133], Avocado e-mail [76, 118], and AITA/WORKSM [71, 112], employ metrics of retrieval performance, including MAP, MRR, and NDCG, as the definitive standard for evaluation. This progression reflects a fundamental principle: the preservation of intent must ultimately enhance the quality of ranking rather than merely maintain linguistic similarity.

Table 5. Dataset and Metrics Comparison in Query Rewriting and Expansion

Field	Dataset	Domain/Source	Personalized Signal	Metrics
Query Rewriting	SCAN [75, 226]	Synthetic commands	Weak (lack personalized feature)	Accuracy
	Robust04 [82]	News corpus retrieval	Medium (no direct personalized feature)	Accuracy
	Avocado e-mail [76, 118]	Enterprise e-mail threads	Strong	BLEU, ROUGE
	Amazon Review [76, 116]	E-commerce reviews (no direct personalized feature)	Medium (no direct personalized feature)	BLEU, ROUGE
	Reddit Comments [76, 155]	Social comments	Medium (no direct personalized feature)	BLEU, ROUGE
	Amazon ESCI [115, 131]	Shopping query-product relevance	Medium (no direct personalized feature)	EM, ROUGE-L, XEntropy
	AOL Query Log [121, 227]	Web search query log	Strong	MAP, MRR, P@1
	WARRIORS [133]	LLM-simulated web sessions	Strong	MRR, NDCG
	AITA and WORKSM [71, 112]	Reddit + workplace posts	Strong	Macro-F1, BS-F1
	PIP	Text-to-image prompts	Strong	PMS, Image-Align, ROUGE-L
Query Expansion	PR2 [9]	Personalized re-ranking	Strong	MAP, MRR, NDCG, RBP
	Delicious [12, 18]	Social bookmarking	Strong	Precision, MAP, MRR, Recall
	Flickr [12]	Photo tagging	Strong	Precision
	CiteULike [13, 17]	Academic bookmarking	Strong	Recall, MAP, MRR
	LRDP [15]	Media preferences	Medium (no direct personalized feature)	Precision, Recall, F1
	BibSonomy [111]	Literature tags	Strong	MAP, Precision, PQEC, $\text{Prof}_{\text{overlap}}$
	Wikipedia [11, 43]	Knowledge graph	Strong	Precision, Recall

Table 6. Dataset and Metrics Comparison in Retrieval

Field	Dataset	Domain/Source	Personalized Signal	Metrics
Retrieval	TOPDIAL [167]	Synthetic role-play	Strong	BLEU, P-F1, K-F1, Succ. (%)
	LiveChat [44]	Chats History	Strong	Recall@k, MRR
	PersonalityEvd [156]	Chinese TV series	Medium (Scripted characters, not authentic)	Accuracy, Human Eval
	Pchatbot [127]	Weibo logs	Strong	BLEU, ROUGE, MRR
	DuLeMon [192]	Chat History	Strong	PPL, BLEU Acc, Precision, Recall, F1
	PersonalityEdit [104]	Synthetic opinion QA	Strong	ES, DD, Acc

5.2 Retrieval

From a retrieval perspective, six primary datasets serve distinct personalization objectives, as detailed in Table 6. TOPDIAL [167] provides target-oriented dialogues suited for goal-aware retrieval by indexing action-topic pairs and preference hints. Similarly, LiveChat [44] enables recipient-aware retrieval in multi-party settings through addressee edges and reply chains. PersonalityEvd [156] offers utterance-level evidence for explainable personality judgments, although the moderate personalization strength of this dataset reflects its scripted nature. Conversely, Pchatbot [127] aggregates authentic social media exchanges for history-centric personalization based on topical proximity and author identity. DuLeMon [192] explicitly encodes mutual persona knowledge and long-term memory to support sophisticated memory-store retrieval cycles, whereas PersonalityEdit [104] functions as a bank of style exemplars for retrieving few-shot examples that match specific trait-topic combinations. Despite these complementary strengths, these datasets exhibit inherent limitations. Notable constraints include the emphasis on goals rather than deep personalization in TOPDIAL, the noise from multi-party interactions in LiveChat, the spurious correlations in Pchatbot, and the synthetic nature of PersonalityEdit.

5.3 Generation

Table 7 presents the datasets and metrics related to generation. LaMP [139] leverages the interaction histories of users across diverse domains for personalized generation, evaluating performance through accuracy, F1, and ROUGE metrics. Similarly, LongLaMP [74] facilitates generation with significantly extended contexts, utilizing the same metrics for evaluation. PGraphRAG [8] extends this paradigm by requiring graph-based retrieval over the interaction structures of e-commerce, and

Table 7. Dataset and Metrics Comparison in Generation

Field	Benchmark/Dataset	Domain/Source	Personalized Signal	Metrics
Generation	LaMP [139]	Multi-domain personalized task	Strong	Accuracy, F1, MAE, RMSE, ROUGE
	LongLaMP [74]	Long-horizon personalized tasks	Strong	Accuracy, F1, MAE, RMSE, ROUGE
	DuLeMon [192]	Open-domain dialogue	Strong	PPL, BLEU, DISTINCT, F1
	PGraphRAG [8]	E-commerce interaction graphs	Strong	ROUGE, METEOR, MAE, RMSE
	AmazonQA/Products [36]	Product QA	Strong	ROUGE, Persona F1
	LoCoMo [102]	Conversations	Strong	QA F1, R@k, FactScore/ROUGE-L, NLG metrics

it employs MAE and RMSE for estimation alongside ROUGE and METEOR scores. Furthermore, DuLeMon [192] and LoCoMo [102] constitute two dialogue datasets that necessitate cross-session continuity. These datasets apply hierarchical summarization or event-based indexing to maintain manageable context windows while preserving temporal causality. Specifically, DuLeMon assesses the quality of dialogue via perplexity, BLEU, and DISTINCT metrics, whereas LoCoMo combines QA measures, such as R@k and FactScore, with comprehensive metrics for natural language generation.

5.4 Agent

Personalized agents have emerged as a significant research focus, designed to adapt to individual signals, contexts, and enduring identities. Table 8 details relevant datasets. In healthcare and physical world applications, these agents leverage physiological data like ECG patterns from the MIT-BIH Arrhythmia database [190] and enable personalized clinical dialogue [2]. Conversely, in social settings, agents maintain specified personas to facilitate natural interaction. To evaluate persona consistency, the research community developed benchmarks including RoleBench [179], Character-LLM [145], InCharacter [173], SocialBench [22], MMRole-Data [34], RolePersonality [130], and CharacterEval [160]. Collectively, these assess model effectiveness in maintaining styles, goals, and social behaviors. Furthermore, advances like Neeko [198] and memory-augmented retrieval [177] allow agents to efficiently adapt identities and update memories. Beyond conversation, personalization extends to action-oriented tasks, encompassing mobile interface control [163], simulated societies with persistent memory like Smallville [120], preference-aware user simulation [2], tailored travel planning [150], and profile-based web navigation [20]. In embodied environments, agents demonstrate grounded personalization by acquiring long-horizon skills in Minecraft [164], following instructions in ALFRED [205], and zero-shot task planning [57]. Simultaneously, OKR-driven frameworks align agent behavior with organizational structures [223]. Finally, knowledge-grounded approaches advance personalization through source-aware dialogue [51], reflective learning from execution traces [134], personality preservation under compression [174], and long-context evaluation like LoCoMo [102]. These frameworks emphasize that durable, retrievable memories are essential for maintaining consistent identity across diverse tasks and contexts.

6 Challenges and Future Directions

Despite recent advancements in Personalized RAG and agent-based systems, several critical challenges and future directions remain that warrant further exploration. We have summarized these into the following six categories, which are detailed below.

6.1 Evaluation of Personalized System

Assessing the degree of personalization in RAG and agent-based systems remains a substantial challenge. Conventional metrics such as BLEU and ROUGE [98, 114, 165] fall short in capturing alignment with individual user preferences. From a personalization standpoint, two users may

Table 8. Dataset and Metrics Comparison in Agent

Field	Dataset Name	Agent Usage	Dataset Source	Application Area
Agent	MIT-BIH Arrhythmia [190]	Physiological signal understanding; real-world sensing demos	PhysioNet + self-collected	Healthcare signals/physical world
	Abbasian et al. [2]	Personalized health dialogue agent	–	Healthcare dialogue
	RoleBench [179]	Train/evaluate role-playing ability	Author-released	Role-playing dialogue
	Character-LLM data [145]	Train agents with character experiences/memory	Author-released	Role-playing dialogue
	InCharacter [173]	Interview dialogues for personality fidelity	Author-released	Personality evaluation
	SocialBench [22]	Evaluate sociality/interaction skills	Author-released	Social conversational ability
	MMRole-Data [34]	Train/evaluate text–image role agents	Author-released	Multimodal role-playing
	RolePersonality [130]	Personalized dialog finetuning	Author-released	Role-playing
	Wang et al. [163]	Conversational control over mobile UI	–	Mobile UI manipulation
	CharacterEval [160]	Evaluate Chinese role-playing agents	Author-released	Chinese role-playing dialogue
	Yu et al. [198]	Efficient multi-character adaptation	–	Role-playing dialogue
	Wang et al. [177]	Personalization memory graphs	Private	Personal assistant/enterprise
	Smallville [120]	Agent memory/behavior evaluation	Author-released	Social simulation/HCI
	Sim-User [2]	Simulate user behavior/conversations	Author-released	User simulation
	Singh et al. [150]	Personalized itinerary planning	–	Travel planning
	PersonalWAB [20]	Train/evaluate personalized web agents	Author-released	Web task agents
	Minecraft Wang et al. [164]	Open-ended skill exploration	Open environment	Embodied games
	ALFRED environment [205]	LLM-guided skill transfer	Public benchmarks (varied)	Problem solving/skill transfer
	Huang et al. [57]	High-level tasks plans	Public	Embodied task agent
	Zheng et al. [223]	Goal-driven multi-agent collaboration	–	Organizational/management agents
	KBP [51]	Persona-grounded, source-aware dialogue	Author-released	Personalized knowledge dialogue
	Renze and Guven [134]	Compare self-reflection strategies for solving problems	Author-released	Reasoning/problem solving
	Wang et al. [174]	Personality consistency Evaluation	–	Role-playing/personality
	LoCoMo [102]	Evaluate long-term conversational memory	Author-released	Long-term conversational memory

exhibit different personalized responses for the same semantic intention, rendering one-size-fits-all metrics potentially misleading. Although human evaluation may be a feasible alternative, it incurs high labor costs and introduces subjective biases. Accordingly, there is an urgent need to develop user-specific evaluation metrics and frameworks that accurately reflect personalized user satisfaction, especially gauging the level of personalization. Additionally, the use of LLMs as judges (LLM-as-Judge) has shown promise for personalization evaluation and warrants increased attention [37].

Another critical limitation lies in the lack of dedicated benchmarks for systematically evaluating personalization across the entire RAG pipeline, including the pre-retrieval, retrieval, and generation phases. Given that current benchmarks mainly focus on single stage, such as LaMP [139] for personalized generation and Dulemon [192] for personalized retrieval, there is an essential need to develop a comprehensive benchmark that quantify the degree of personalization across various stages in RAG, as well as evaluate how these stages interact to shape the overall personalized generation. Future work should focus on designing such quantitative frameworks to enable rigorous and holistic personalization assessment.

6.2 Preserving Privacy through Edge-Cloud Collaboration

Personalization often entails handling sensitive personal data (e.g., search history, location, health information), raising serious privacy concerns. With regulations like the EU's GDPR enforcing strict controls on personal data usage, personalized RAG systems must be designed to protect user information by default. A central challenge is that the conventional approach of sending all user queries and context to a central LLM may violate privacy or data locality requirements. Moreover, storing rich user profiles on servers risks breaches or misuse. To comply with privacy-preserving regulations, personal data should not be transmitted or stored by the system. Thus, a promising

direction is to preserve user-sensitive features within edge devices while delegating computation-intensive tasks to cloud-based systems. In this paradigm, typically a smaller model on the user's device handles sensitive, user-specific tasks, leveraging personal data locally, while a powerful cloud LLM provides general reasoning and knowledge, constrained to use only anonymized or high-level signals from the device [117], ensuring that raw personal data never leaves the user's device, significantly mitigating privacy risks. Recent works like FDLoRA [126], a personalized federated learning framework, train local adapter modules on each user's data and only upload aggregated insights to the server. Overall, by leveraging on-device models, federated updates, and encryption, future personalized agents can satisfy stringent privacy regulations, proving that effective personalization need not come at the expense of user privacy.

6.3 Scalability and Efficiency

Scalability is one of the critical challenges in personalized systems, since current systems must efficiently manage large user bases, each with unique personalization attributes. As the number of users and the complexity of their personalized data increase, balancing efficiency and scalability becomes a critical concern. To address scalability in personalized RAG systems, both retriever and generator components can be optimized [64]. For retrievers, designing lightweight architectures [178] or employing user grouping strategies [106] can enhance retrieval efficiency. In the generation phase, parameter-efficient tuning methods offer a lightweight approach to model fine-tuning, enabling improved personalization with minimal overhead [157, 159]. However, as discussed in Section 4.3, while techniques like LoRA can enhance personalization, the cost associated with individual user fine-tuning becomes prohibitive as the user base grows. Therefore, developing lightweight, multi-user LoRA fine-tuning frameworks represents a promising direction for future research. Another scalability issue is the cold-start problem, where the agent system struggles with task performance for new users or tasks, or where agent systems rely only on predefined rules or toolsets that cannot generalize to new tools, thereby limiting emergent behaviors [4]. Addressing the cold-start problem necessitates meta-learning and continual adaptation capabilities, enabling agents to learn dynamically from minimal feedback. A promising research avenue involves developing meta-reasoning mechanisms that dynamically determine when to leverage general knowledge, past interactions, or new tools based on confidence levels [6]. These mechanisms would allow agents to rapidly personalize strategies for unseen situations, thereby enhancing scalability across diverse users and environments.

6.4 Ethical and Coherent System

The pursuit of personalization must be carefully balanced with robustness and ethical considerations to foster user trust. A primary concern is that personalization may inadvertently introduce or amplify biases. When underlying data or inferred user profiles contain biased attributes (e.g., gender, ethnicity), personalized models risk reinforcing stereotypes by selectively exposing certain demographics to particular content. Recent studies on personalized generation caution that such systems may produce discriminatory outputs if not properly monitored [194]. There is a moral obligation to ensure that personalization enhances the user experience without imposing unfair constraints or exposing personal information. Recently, researchers advocate for the integration of fairness audits and bias mitigation strategies into personalized intelligence systems [42, 54]. These approaches include debiasing personalized recommendations, employing adversarial training to eliminate signals from protected attributes, and enabling users to inspect and modify the profile data that the system uses.

Beyond fairness, maintaining coherence and reliability throughout the RAG pipeline presents a critical technical challenge. Personalized systems often function end-to-end, retrieving context

before generating responses. Misalignments between these stages can lead to errors, such as when the retrieval module surfaces relevant personal facts that the generation module subsequently ignores or contradicts through hallucinations. Ensuring that model reasoning remains grounded in retrieved evidence and user context is thus essential. Recent research identifies the divergence between a LLM's reasoning trajectory and the retrieved evidence as a central failure mode in RAG systems [180]. To mitigate these inconsistencies, researchers propose joint optimization and feedback mechanisms, such as AlignRAG [180], which incorporates an iterative critique step during generation to enforce alignment with retrieved content, thereby correcting reasoning drift. Complementary approaches leverage persistent memory modules that retain user-specific information, reducing hallucinations and preserving consistency with the user's persona or history [2].

6.5 Multimodal Personalization

Most personalized RAG efforts remain text-centric, yet personal assistants increasingly interact through images, audio, video, and interface traces. Multimodal LLMs enable modeling users not only linguistically but also visually and acoustically. Recent work at the intersection of personalization and multimodality has focused on personalized generation [147, 172, 193, 195]. These approaches distill user history into factorized, editable preference slots and drive multimodal generators with those slots to generate user preference-aligned content. Multimodal RAG extends this paradigm by fetching user-owned artifacts such as style boards, reference photos, and brand kits, while providing cross-modal constraints that enrich the available multimodal resources. Works like ImageRAG [144] and Multi-RAG [103] advocate dynamic retrieval of images, video, and audio to stabilize generation and reasoning under sparse training coverage, particularly for long-tail concepts and streaming contexts. Emerging personalized multimodal-aware agents further expand application scenarios from text-based to action-based interactions, including AppAgent for application navigation [202], Pathfinder for multimodal diagnostic agents [46], and EmbodiedBench [196], which provides vision-driven embodied agents across four environments ranging from high-level semantic tasks in household settings to low-level atomic actions such as navigation and manipulation.

Despite these advances, multimodal RAG and agent systems face several critical challenges that require attention. First, retrieval granularity remains a significant issue, necessitating a shift from document-level to patch-, segment-, or track-level retrieval, including image patches, audio timbre segments, and video shot graphs, which better aligns with how preferences manifest across modalities. Current work lacks comprehensive approaches to personalized hybrid multimodal retrieval. Second, there is a need to train multimodal preference critics capable of cross-modal reasoning between different modalities to detect violations such as incorrect color palettes or identity drift for personalization. Third, safety and calibration in multimodal information fusion present substantial concerns, as personalization must remain truthful and fair by coupling preference control with safety constraints, identity-preservation checks, and calibrated uncertainty, privileging retrieved facts over inferred preferences when conflicts arise. Fourth, developing lifelong, privacy-preserving personalized agents which are truly embedded into human life is personalization's final goal of serving the community.

6.6 Personalized Agent Planning

Recently, autonomous agents powered by LLMs (LLMs) have demonstrated remarkable capabilities in reasoning, tool use, and executing complex, multi-step tasks [176, 208–210]. However, extending personalization into the planning and decision-making part of these autonomous agents remains a largely uncharted area. Current LLM-based agents primarily focus on general capabilities (e.g., conversation, GUI navigation), with little attention to tailoring their plans to individual users. Consequently, today's agent frameworks often treat all users the same. There are early works that

apply them in specialized domains: in healthcare, Conversational Health Agents [3] are proposed to provide personalized healthcare assistance and diagnosis; in education, ExpertAgent [228] is proposed to provide personalized learning content and strategies; in e-commerce, extending traditional item recommendation strategies [61, 63, 72, 83, 95, 96, 99, 219–222] design conversational sales agents. Such examples underscore the potential benefits of personalized planning, but they remain isolated cases. By and large, the integration of personalization into agent planning algorithms (the logic that decides what steps to take next) is still in its infancy and not part of most agentic systems or benchmarks. Embedding personalized decision-making in agent frameworks is a compelling future direction. One approach is to design a long-term memory module to optimize its choices. Preliminary work like RecAgent [170] augments an LLM-based agent with a hierarchical memory of the user's past interactions, allowing it to plan recommendations across multiple domains in a user-tailored way. TravelPlanner [188] uses memory-augmented LLM planning to incorporate a user's travel history and preferences when building a travel itinerary, rather than planning in a one-size-fits-all manner. Building on these ideas, future agents could develop more human-aligned and dynamically evolving memory systems to adapt to agent interactions across multiple scenarios. Other mechanisms like meta-learning [33, 171], where an agent learns to plan differently for different user profiles, or constraint-based planning that respects user-specific rules, offer promising directions. Success in this area would enable truly personalized agents, creating systems that not only retrieve and generate information for the user but also autonomously act and plan in ways that align with that user's unique needs.

7 Conclusion

In this article, we explore the landscape of personalization from RAG to advanced LLM-based Agents, detailing adaptations across pre-retrieval, retrieval, and generation stages while extending into agentic capabilities. By reviewing recent literature, datasets, and metrics, we highlight the progress and diversity in enhancing user satisfaction through tailored AI systems.

However, critical challenges remain that require prioritized attention to advance personalization systems effectively. Chief among these is the lack of robust evaluation frameworks, as current research lacks standardized benchmarks and personalization-specific metrics capable of accurately measuring real user satisfaction and system performance. Three fundamental technical and operational challenges follow: privacy protection, scalability constraints, and ethical considerations. Addressing these interconnected issues is essential to building more trustworthy and effective RAG and agentic systems that deliver reliable, user-centered services. Looking forward, multi-modal integration and enhanced agent planning capabilities represent promising directions for achieving higher-level, more human-aligned personalization. To support ongoing progress in these critical areas, we maintain a continuously updated repository of relevant papers and resources.

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